

Gradebook Instruction Manual

Overview:

This software application is designed to help educators and students manage academic performance efficiently. It simplifies the tracking of student grades, facilitates the management of courses, and generates useful statistics for both educators and students. The application is a part of an Educational Resource Planning (ERP) system we made to enhance the educational experience.

Libraries Used:

tkinter: provides the graphical user interface (GUI)

sqlite3: manages database operations for storing course and student data

string: handles string manipulations

os: deals with file operations (such as checking for the database)

The rest of the libraries needed are a part of Python's standard library

Features of Product:

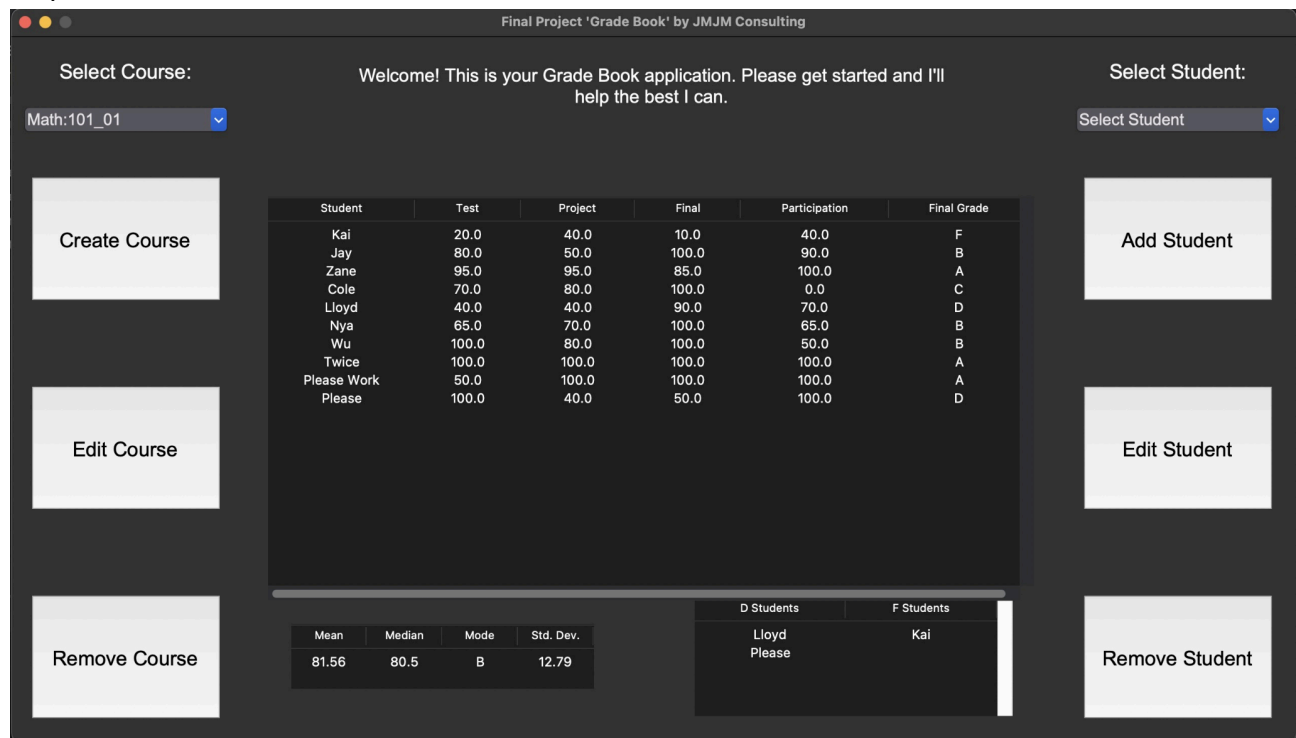
1. Course Management
 - a. Create, edit, and delete courses
 - b. Assign students to courses
2. Student Management
 - a. Add, edit, and delete student records
 - b. Track individual student grades
 - c. Display students who are at risk of failing based on their grades
3. Gradebook System
 - a. Provides an easy way for educators to input grades and view class statistics
 - b. Displays student grades and identifies those who are close or failing in courses
4. Courses Statistics
 - a. Provides both educators and students with course statistics, such as average grade, highest grade, and lowest grade

How to Use Application:

Step 1: Start the Application

- Run the Python file (add the file) to launch the application
- Make sure that Python is installed on your computer, since this code relies on some of Python's default libraries

Step 2: Main Interface



Main Window: Displays a list of courses currently in the system and you can use the following options

- Create Course: Opens a window to input new course details
- Edit Course: Allows modification of a selected course's details
- Remove Course: Removes a selected course from the database
- Add Student: Adds a new student to a selected course
- Edit Student: Modifies a student's details in a selected course
- Remove Student: Removes a selected student from the database

Detailed Walkthrough:

Create A New Course:

Please enter in everything, be sure the weights total to 100%, and have grade scale values be logical!

What is the course NAME?

What is the course NUMBER?

What is the course SECTION?

How many STUDENTS are there?

Accept

Cancel

Class Components:

Component Name	Weight (%)
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Enter Course Component:

Add

Enter Respective Weight:

Change

Selected Component From Table:

Delete

Grade Scale:

A >=

B >=

C >=

D >=

F <

Please get started and I'll help the best I can.

1. Creating a Course

- Click the **Create Course** button on the main interface
- A new window will open asking for details such as course name, course code, and other information
- After filling out the details, click **Save** to add the course or **Cancel** to close without saving

Edit A Current Course:

You are currently editing the following course: French:200_01

What is the course NAME?

French

What is the course NUMBER?

200

What is the course SECTION?

01

You can change the STUDENTS count manually in the main window!

Save

Cancel

Class Components:

Component Name	Weight (%)
Test	10
Test 2	20
Group Project	70

Enter Class Component (Name):

Add

Enter Respective Weight (1-100):

Change

Selected Component From Table:

Delete

Grade Scale:

A >=

90

B >=

80

C >=

60

D >=

50

F <

50

Please get started and I'll help the best I can.

2. Editing a Course

- Select a course from the list and click **Edit Course**
- A window will open with the course's current details. Modify the information if needed.
- Click **Save** to save changes or **Cancel** to discard changes.

3. Deleting a Course

- Select a course and click **Delete Course**
- A confirmation box will appear to confirm the deletion. Click **Yes** to delete the course or **Cancel** to exit without deleting it.

Add a New Student

Please enter in as much as you can according to the new student's current status!

What is the Student's NAME?:

Add

Cancel

Components	Grades
t	100.0

Select a Component to start Grading!

Apply

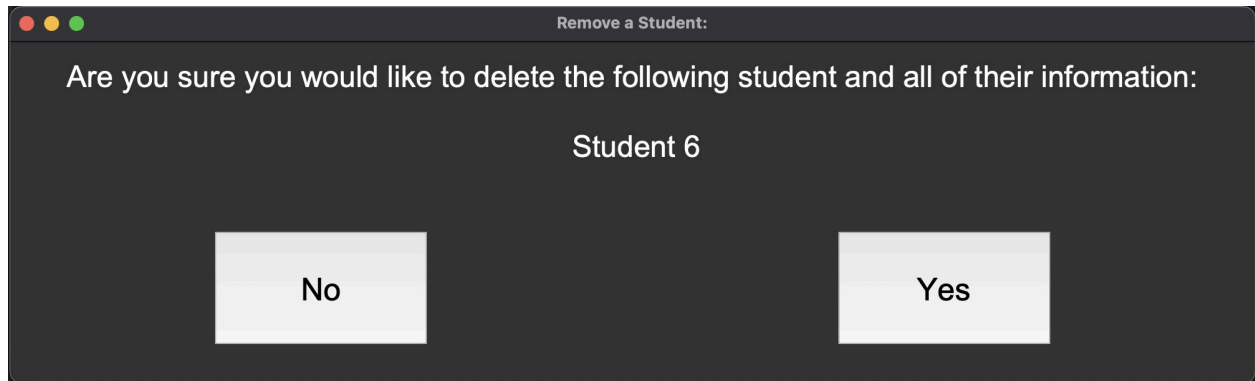
Please get started and I'll help the best I can.

4. Adding a Student

- Select the course to which you want to add a student from the main interface
- Click the **Add Student** button
- A window will open where you can input the student's details
- Click **Save** to add the student or **Cancel** to exit without saving

5. Editing a Student

- Select the course and then select the student you wish to edit
- Click the **Edit Student** button
- Modify the student's information as needed. Click **Save** to confirm changes, or **Cancel** to discard changes.



6. Deleting a Student

- Select a student and click **Delete Student**
- A confirmation box will appear to confirm the deletion. Click **Yes** to delete the course or **No** to exit without deleting it.